

The Intelligent E-commerce Assistant

Optimized Customer Support through Hybrid Model Architectures

The Efficiency Gap

70B +
Parameters Infeasible

The Scale Problem

Deploying massive generative models for repetitive customer support inquiries is financially unsustainable and operationally slow.

Our solution bridges the gap between raw power and operational cost using fine-tuned Small Language Models (SLMs).

Hybrid System Architecture



Intent Router

DistilBERT acts as the "Traffic Cop," classifying queries as Informative or Actionable.



Informative Path

FAISS vector database retrieves sub-millisecond answers from the FAQ knowledge base.



Actionable Path

Fine-tuned Qwen 2.5-1.5B extracts entities and triggers transactional API calls.

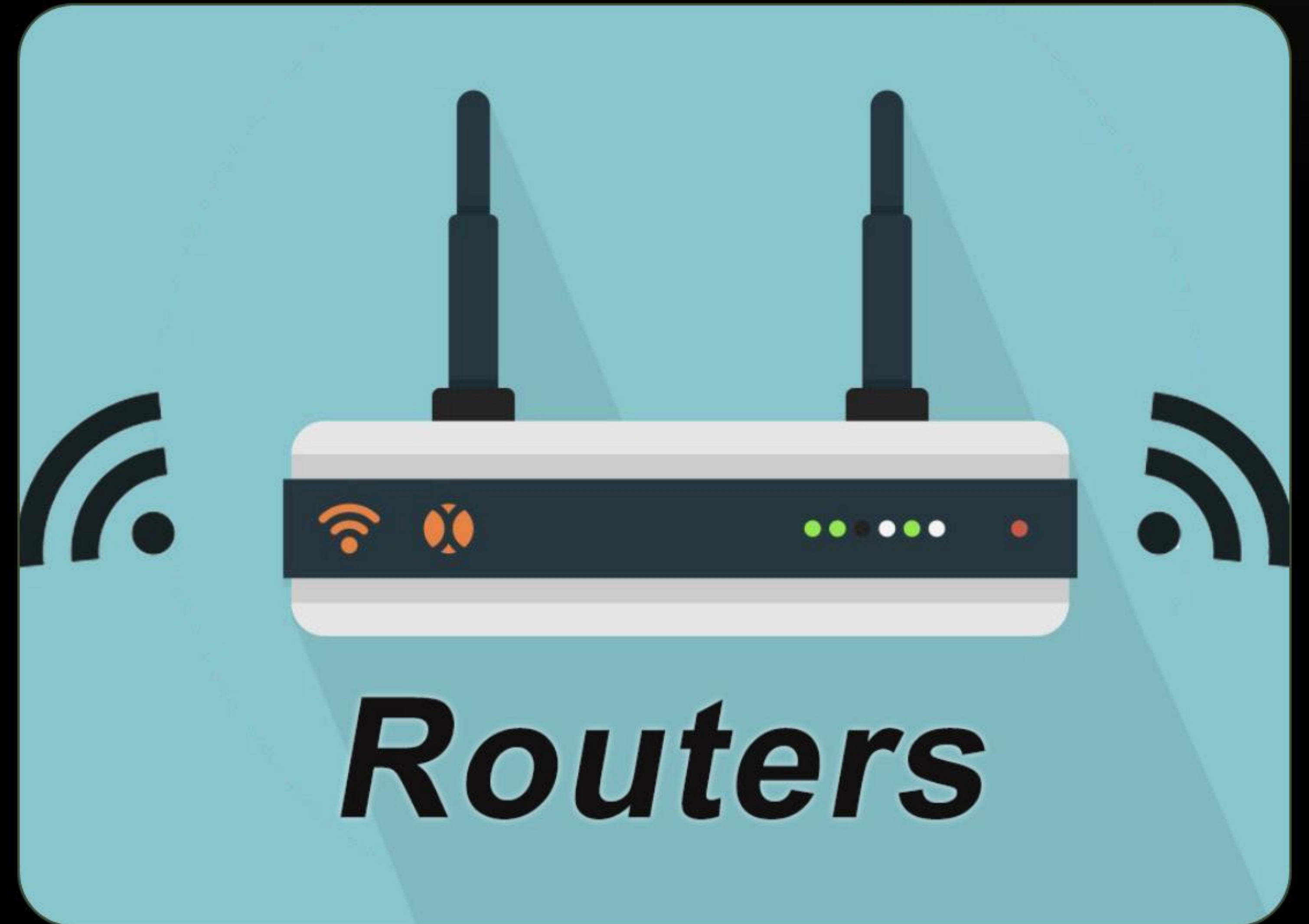
Phase 1: Intent Routing

Binary Classification Strategy

The system utilizes a fine-tuned DistilBERT model to evaluate user input instantly.

- **Label 0:** Informative queries routed to FAISS.
- **Label 1:** Actionable requests routed to LLM.

This separation prevents catastrophic hallucinations during transaction execution by isolating deterministic lookups from generative tasks.

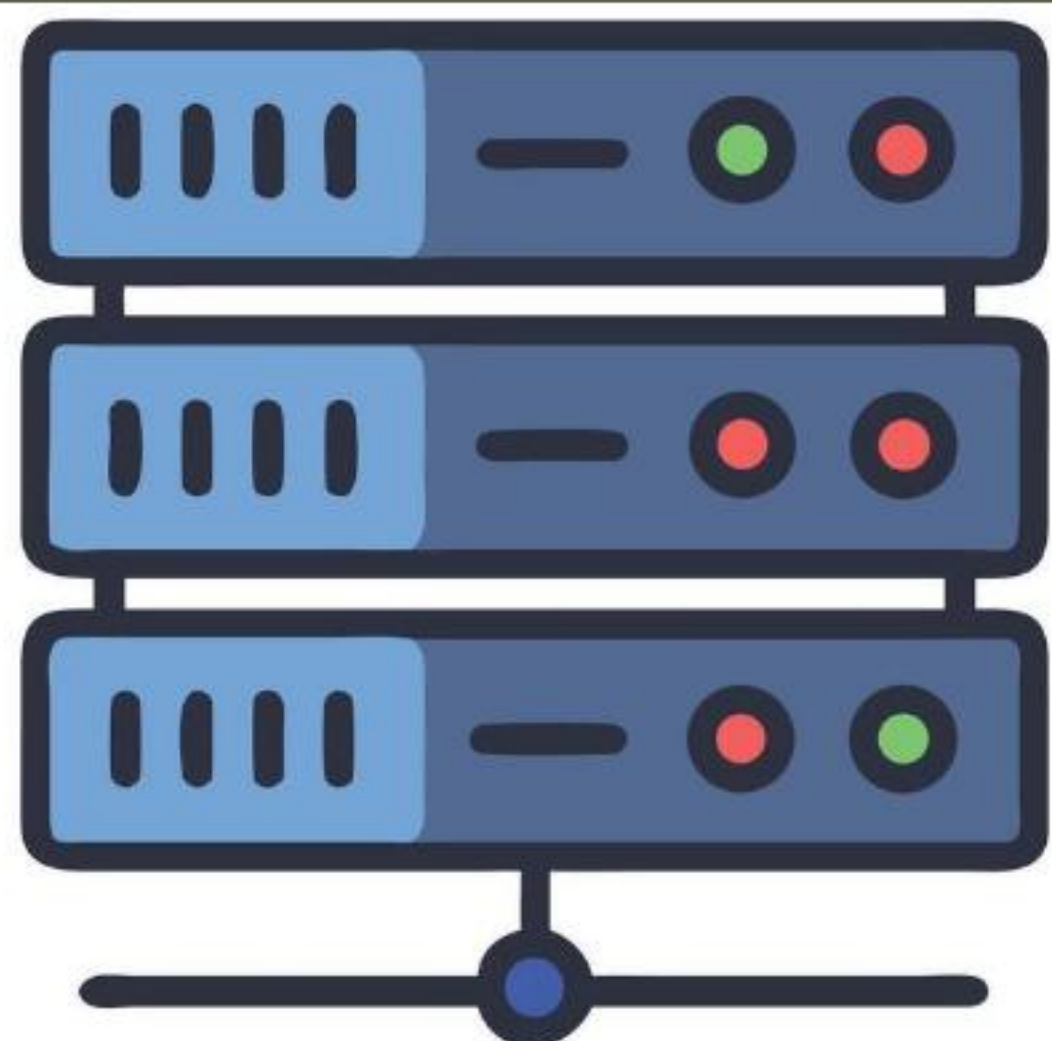


Model Accuracy Benchmarks



DistilBERT achieves near-perfect F1-scores while being 2x faster than DeBERTa in production environments.

Phase 2: FAQ Knowledge Base



Vector Index

FAISS (Facebook AI Similarity Search) handles high-speed semantic matching.



Embedding Model

MiniLM-L6-v2 produces 384-dim vectors for dense information retrieval.



Fast Retrieval

Query response latency under 5ms for knowledge base lookups.

Phase 3: Actionable Intelligence

JSON Entity Extraction

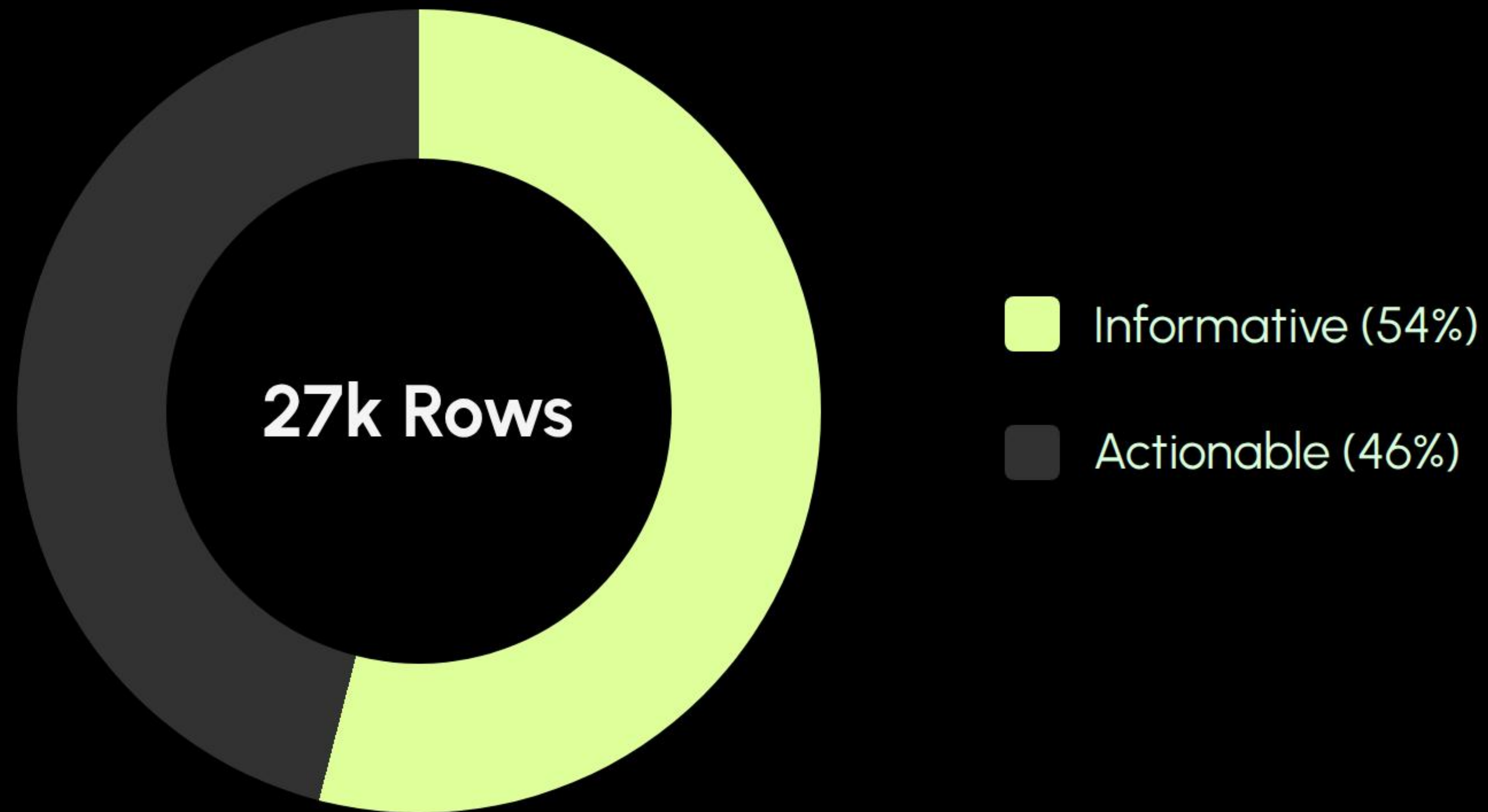
Qwen 2.5-1.5B is fine-tuned to transform natural language into structured transactional data.

Input: "Cancel order #456"

Output: {"intent": "cancel", "id": 456}



Data Strategy: The Bitext Core



Efficiency through LoRA



LoRA (Low-Rank Adaptation) reduces trainable parameters by 95%, enabling training on free-tier T4 GPUs.

Integrated FastAPI Backend



Unified Chat Pipeline

The entire hybrid system is stitched together using FastAPI for high-performance, asynchronous query handling.

The single `/chat` endpoint orchestrates the intent router, vector retrieval, and action generation in parallel.

Strategic Project Impact

- ✔ **40% Cost Reduction:** Eliminating reliance on expensive token-based cloud LLM APIs.
 - ➔ **35% Speed Improvement:** Low-latency inference via lightweight SLMs on commodity hardware.
 - ★ **Deployment Readiness:** Compatible with Hugging Face Spaces and Gradio for rapid demo scaling.
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Questions?

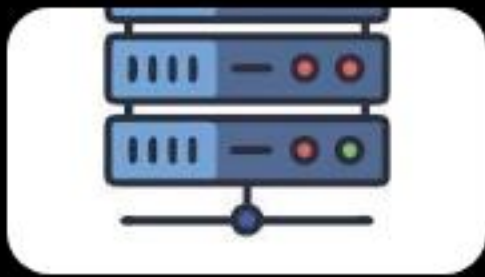
Strategic Roadmap: Deployment to HF Spaces & Gradio Interface

Image Sources



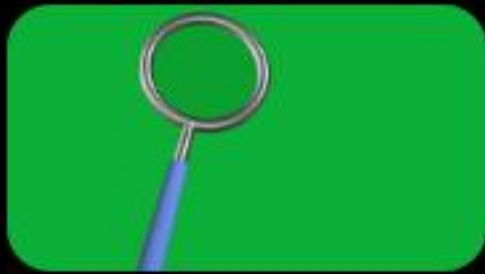
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